SUMMARY · MK 4 OPERATIONS & SUPPLY CHAIN MANAGEMENT · WEEK 7

# Waiting Lines and Simulation

Queues form even when average capacity exceeds demand. Service vs waiting cost, variability, queue structure, the four Jacobs models, Little's Law, explosion at high utilisation, discrete-event simulation, ABM and generative ABM. Case: KAI Commuter at Manggarai.

## Material

### 1 The waiting line problem and variability

Trade service cost (visible) against waiting cost (often unpriceable). Variability drives queues; constant service halves the queue. Kingman: $Wq \approx [\rho/(1-\rho)] \times [(Ca^2 + Cs^2)/2] \times$ service time.

### 2 Anatomy of a queuing system

Five choices: source population (finite/infinite), arrival pattern, queue discipline (FCFS), service configuration (channels parallel, phases in series), exit. Mis-map the structure and the arithmetic is meaningless.

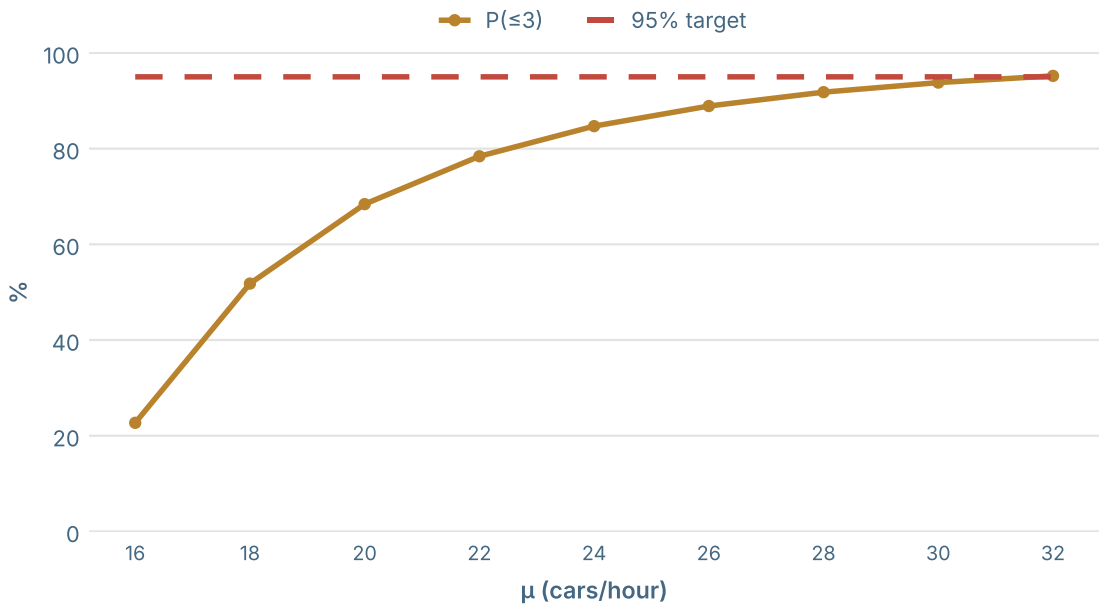
### 3 The four models and formulas

Model 1 (M/M/1): $Lq = \lambda^2/[\mu(\mu-\lambda)]$, $Wq = Lq/\lambda$; Model 2 (constant): half of Model 1; Model 3 multichannel; Model 4 finite population. Little's Law $L = \lambda W$. Western National Bank: $\lambda=15$, $\mu=20 \rightarrow Lq$ 2.25, Wq 9 min; 95% service level needs $\mu \approx 32$. $Lq = \rho^2/(1-\rho)$: 1.63 at 70%, 4.82 at 85%, 18.05 at 95%.

## 4 Queue simulator: feel the explosion yourself Deep dive

Defaults: Western National Bank (λ 15, μ 20 → ρ 75%, Lq 2.25, Wq 9 min).

**P(≤3 cars) vs service rate μ (λ = 15/h)**



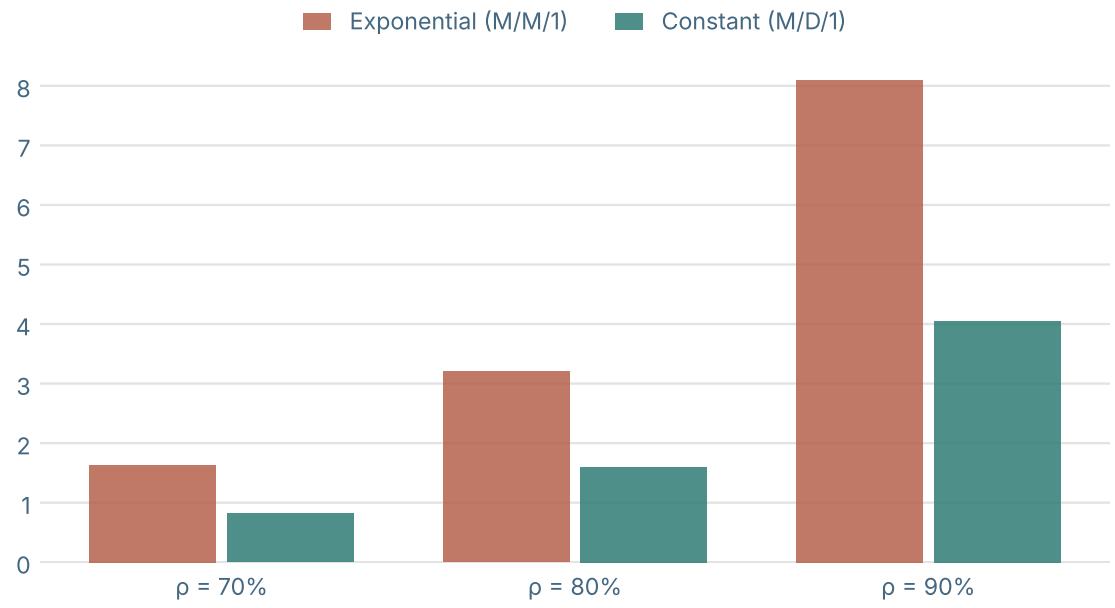

95% confidence of ≤3 cars needs μ ≈ 32/h; the teller idles ~53%.

## 5 When formulas fail: simulation

Formulas assume steady state, Poisson, FCFS, passive customers; reality has balking, reneging, jockeying. Two-stage lines block and starve (Bob 88.7%, Ray 78.2%). DES jumps the clock between events and yields distributions; ABM produces emergent patterns (Schelling); generative ABM uses LLM agents — realistic but hard to validate. Escalate complexity only when forced.

## 6 Variability: constant service halves the queue Deep dive

**Lq: exponential vs constant service**



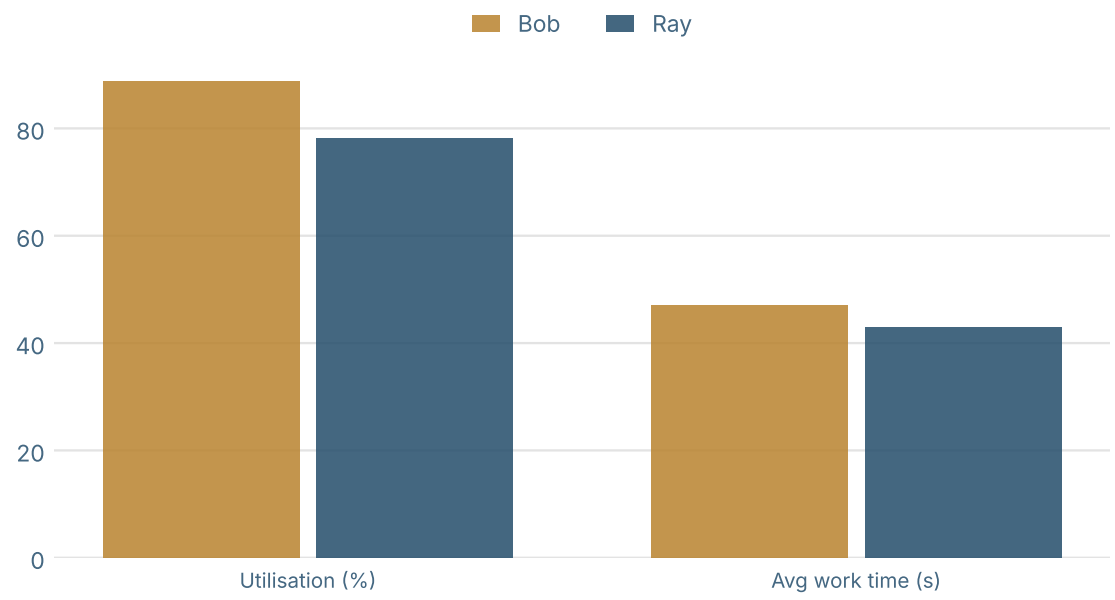

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Kingman: Wq ≈ [ρ/(1−ρ)] × [(Ca² + Cs²)/2] × service time
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Constant service halves the queue at the same utilisation.

## 7 Bob & Ray and Manggarai: when formulas aren't enough

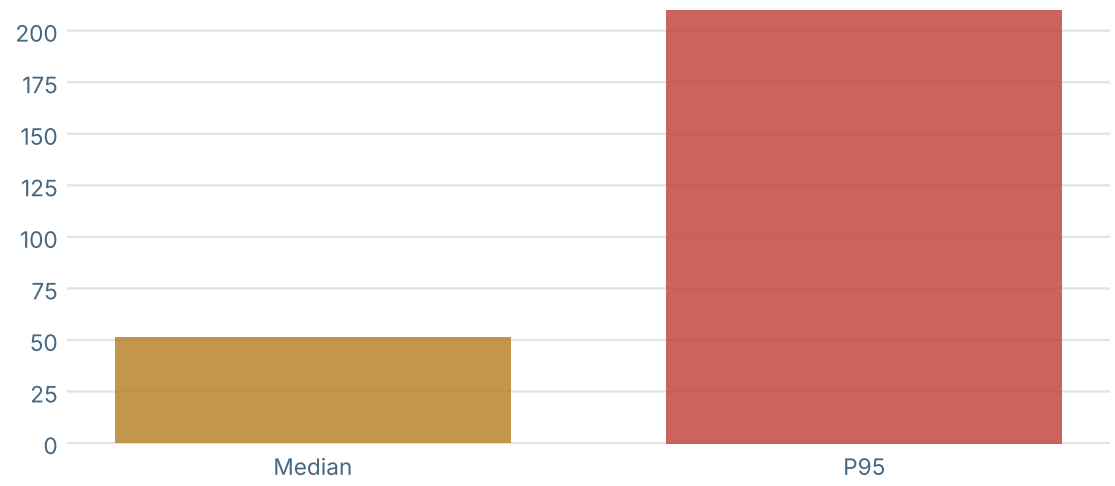
Deep dive

**Bob & Ray simulation (10 units)**




Bob 88.7%, Ray 78.2%: blocking and starving need simulation.

**Manggarai: median vs P95 corridor wait (s)**




Ratio 4.1 vs ~5.4 for M/M/1 at ρ 0.84; arrival waves prove the variability.

| Intervention | Acts on |
|---|---|
| 4 extra trains | Platform batch capacity |

| | |
|---|---|
| Security staff | λ / perception |
| Hall partition | λ |
| Portable counters | S |
| Stop-and-go | λ |

## Key takeaways

1. Variability creates queues.
2. Constant service halves the queue.
3. $L_q = \lambda^2/[\mu(\mu-\lambda)]$.
4. $L = \lambda W$.
5. Queues explode as $\rho \to 1$.
6. Averages hide queues — use P95.
7. Escalate modelling only when forced.

## Case studies

Case & worked answers

### KAI Commuter & the Manggarai capacity dilemma

Four queues in series: gates (Model 2), counters (Model 3), the corridor (flow capacity) and platforms (batch service). Q1: an M/M/1 at ρ=0.84 gives a P95/median ratio ≈ 5.4, so the observed 4.1 alone doesn't prove excess variability; wave arrivals and a corridor empty 60% / jammed 15% do — a variability problem. Q2: four of five interventions act on λ (security, partitioning, stop-and-go, extra trains spreading load); portable counters add S — actions map constraints. Q3: scheduling questions can use existing data; capital questions need DES. Q4: set a service/safety standard as a constraint before costing options.

## Worked examples

**Problem 1** **Medium** **Antrean M/M/1**

Kasir: rata-rata 24 pelanggan datang per jam (λ), kasir melayani 30 per jam (μ). Hitung utilisasi, panjang antrean, waktu tunggu, dan waktu total di sistem.

1. $\rho = \lambda \div \mu = 24 \div 30 = 0{,}8$ (80%).
2. $Lq = \lambda^2 \div [\mu(\mu - \lambda)] = 576 \div (30 \times 6) = 3{,}2$ orang
3. $Wq = Lq \div \lambda = 3{,}2 \div 24 = 0{,}133$ jam = 8 menit.
4. $Ws = 1 \div (\mu - \lambda) = 1 \div 6$ jam = 10 menit (8 menit antre + 2 menit dilayani).
5. Peluang kasir kosong $P_o = 1 - \rho = 20\%$.

**Answer:** ρ 80%, Lq 3,2 orang, Wq 8 menit, Ws 10 menit.

Problem 2 Easy **Little's Law**

Sebuah stasiun dilewati 120 penumpang per jam; tiap penumpang rata-rata berada di stasiun 15 menit. Berapa rata-rata jumlah orang di stasiun?

1. $L = \lambda \times W$
2. Samakan satuan: W = 15 menit = 0,25 jam.
3. L = 120 × 0,25 = 30 orang.

**Answer:** Rata-rata 30 orang di stasiun.

## Practice questions

Try answering first. The answer key is below each question.

**1. λ 15, μ 20. Lq = ...**

A. 0.75
B. 2.25
C. 3
D. 9

**Answer: B.** 225/100.

**2. Little's Law states...**

A. L = λW
B. L = μ − λ
C. W = λ/μ
D. ρ = μ/λ

**Answer: A.** Any stable system.

**3. At equal utilisation, constant service gives a queue…**

A. Same
B. Double
C. Half
D. Zero

**Answer: C.** Half.

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**4. With a small source population (4 machines)…**

A. Model 1
B. Model 2
C. Model 3
D. Model 4

**Answer: D.** Each breakdown changes arrivals.

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**5. Customers leaving because the line is long are…**

A. Jockeying
B. Balking/reneging
C. Batching
D. Blocking

**Answer: B.** Balk vs renege.

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**Sources**


- [STS] Meeting 7 - Waiting Line and Simulation (MK4)
- Jacobs & Chase, Bab 10
- KAI Commuter & Manggarai case
- Schelling (1971), Dynamic Models of Segregation